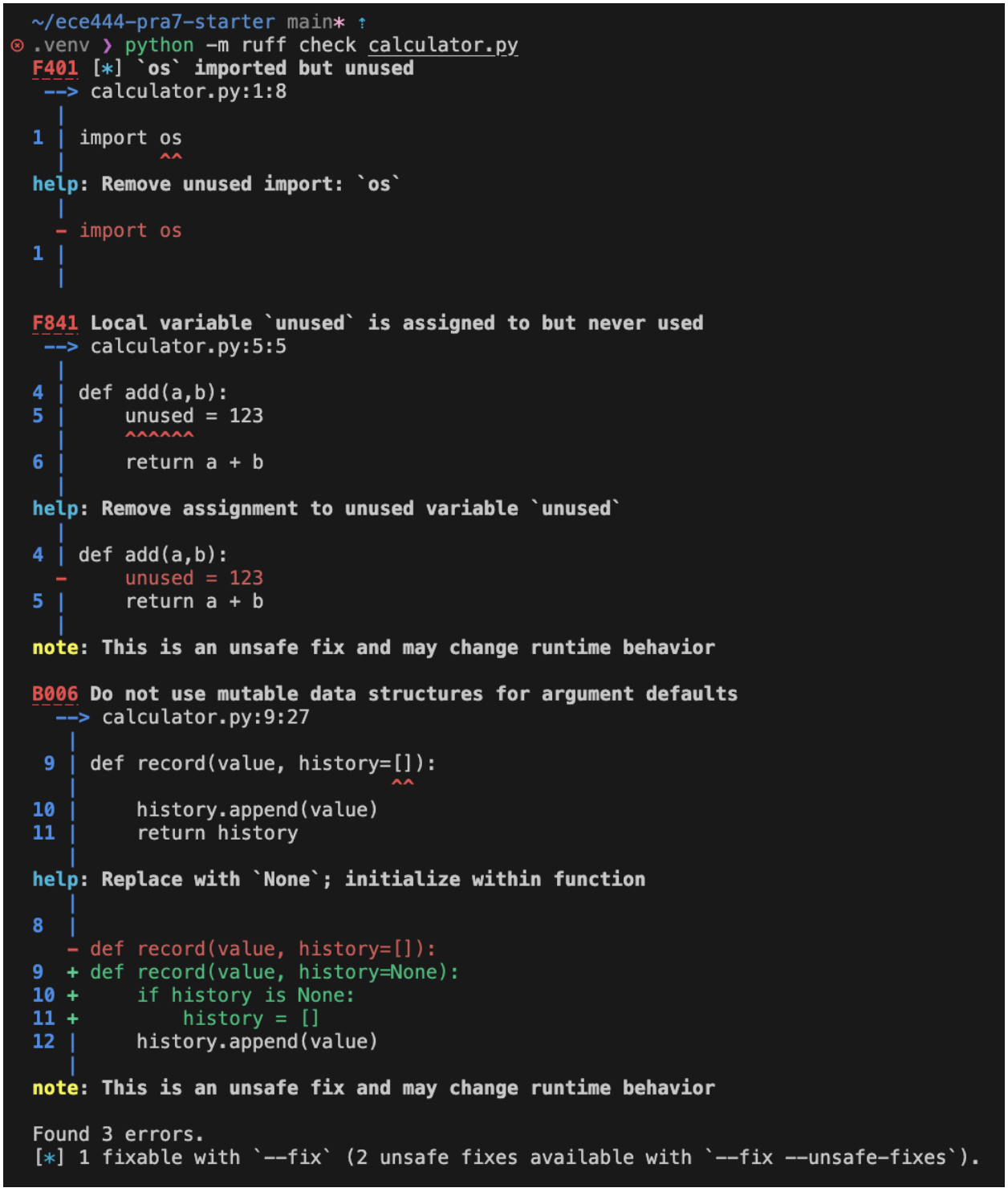
Item 1: Fork repository URL
https://github.com/animalan/ece444-pra7-starter/

Item 2: Ruff findings before fixing



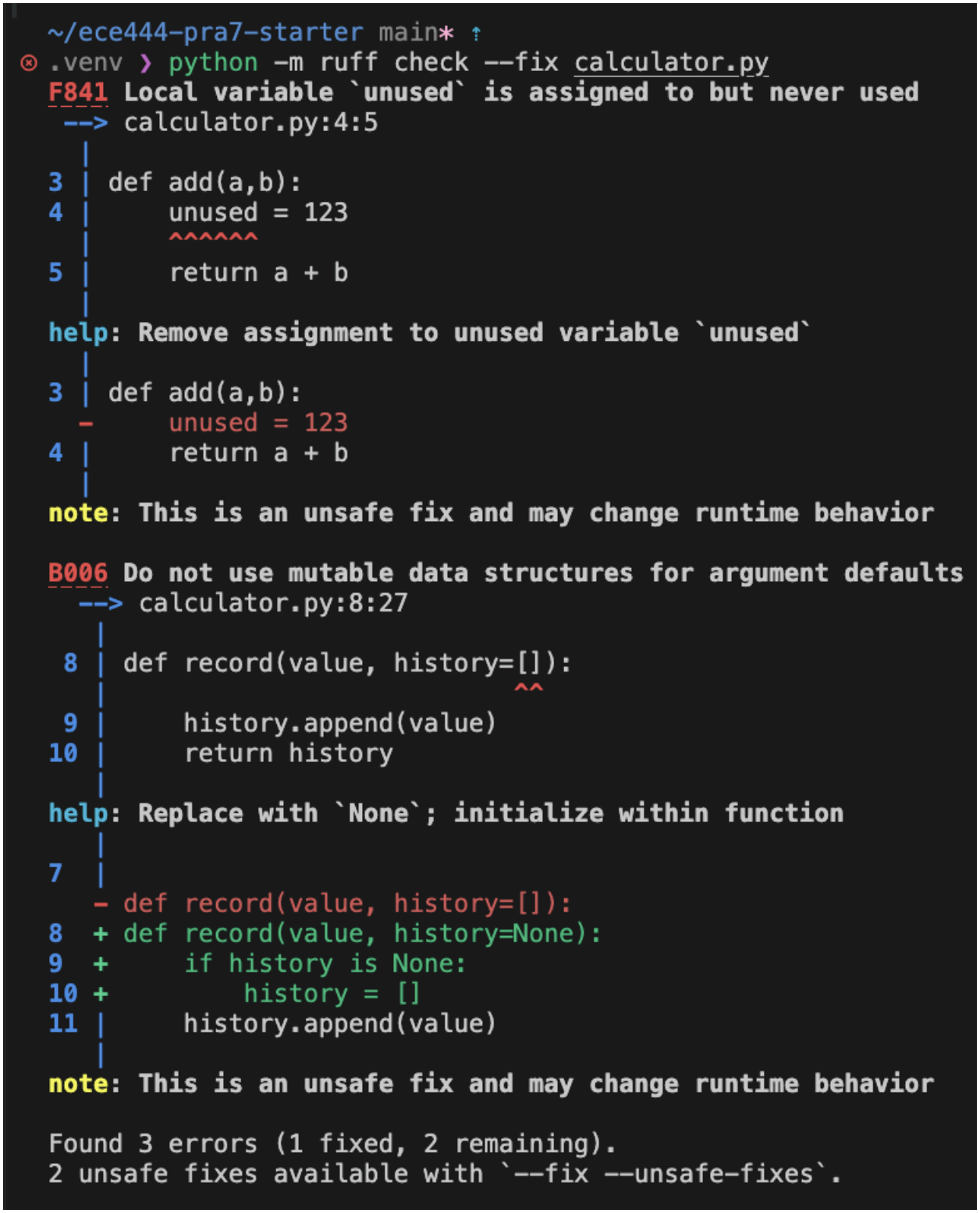
```
~/ece444-pra7-starter main* ⇡
.venv ❯ python -m ruff check calculator.py
F401 [*] `os` imported but unused
 --> calculator.py:1:8
  |
1 | import os
  |        ^^
help: Remove unused import: `os`
  |
  - import os
1 |
  |

F841 Local variable `unused` is assigned to but never used
 --> calculator.py:5:5
  |
4 | def add(a,b):
5 |     unused = 123
  |     ^^^^^^
6 |     return a + b
  |
help: Remove assignment to unused variable `unused`
  |
4 | def add(a,b):
  -     unused = 123
5 |     return a + b
  |
note: This is an unsafe fix and may change runtime behavior

B006 Do not use mutable data structures for argument defaults
  --> calculator.py:9:27
   |
 9 | def record(value, history=[]):
   |                           ^^
10 |     history.append(value)
11 |     return history
   |
help: Replace with `None`; initialize within function
   |
8  |
   - def record(value, history=[]):
9  + def record(value, history=None):
10 +     if history is None:
11 +         history = []
12 |     history.append(value)
   |
note: This is an unsafe fix and may change runtime behavior

Found 3 errors.
[*] 1 fixable with `--fix` (2 unsafe fixes available with `--fix --unsafe-fixes`).
```

Item 3: Ruff automatic fixes



```
~/ece444-pra7-starter main* ⇡
.venv ❯ python -m ruff check --fix calculator.py
F841 Local variable `unused` is assigned to but never used
 --> calculator.py:4:5
  |
3 | def add(a,b):
4 |     unused = 123
  |     ^^^^^^
5 |     return a + b
  |
help: Remove assignment to unused variable `unused`
  |
3 | def add(a,b):
  -     unused = 123
4 |     return a + b
  |
note: This is an unsafe fix and may change runtime behavior

B006 Do not use mutable data structures for argument defaults
  --> calculator.py:8:27
   |
 8 | def record(value, history=[]):
   |                           ^^
 9 |     history.append(value)
10 |     return history
   |
help: Replace with `None`; initialize within function
   |
7  |
   - def record(value, history=[]):
8  + def record(value, history=None):
9  +     if history is None:
10 +         history = []
11 |     history.append(value)
   |
note: This is an unsafe fix and may change runtime behavior

Found 3 errors (1 fixed, 2 remaining).
2 unsafe fixes available with `--fix --unsafe-fixes`.
```

Item 4: Repaired calculator.py

```
  ~/ece444-pra7-starter main* ↑
● .venv ❯ cat calculator.py


  def add(a,b):
      return a + b


  def record(value, history=None):
      if history is None:
          history = []
      history.append(value)
      return history

  ~/ece444-pra7-starter main* ↑
● .venv ❯ git diff
  diff --git a/calculator.py b/calculator.py
  index f6def6c..bc24b12 100644
  --- a/calculator.py
  +++ b/calculator.py
  @@ -1,10 +1,11 @@


   def add(a,b):
  -    unused = 123
       return a + b


  -def record(value, history=[]):
  +def record(value, history=None):
  +    if history is None:
  +        history = []
       history.append(value)
       return history

  ~/ece444-pra7-starter main* ↑
● .venv ❯ python -m ruff check calculator.py
  All checks passed!
```

```python
def add(a,b):
    return a + b

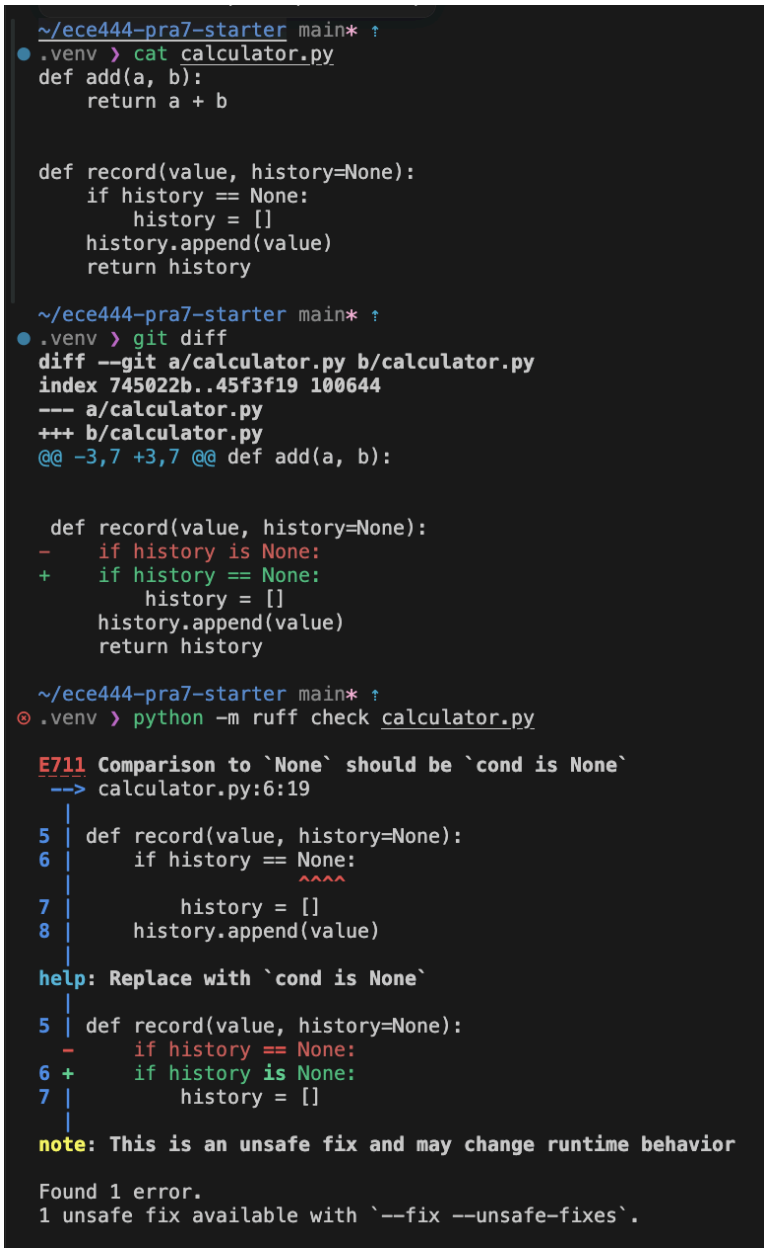


def record(value, history=None):
    if history is None:
        history = []
    history.append(value)
    return history
```

Item 5: Additional violation (before and after)

Explanation: E711 flags comparing to None with ==. None is a singleton and == can be overridden by a class's __eq__, so the correct check is identity. I changed history == None to history is None.

Before (E711 Violation; == instead of is)




```
def add(a, b):
    return a + b


def record(value, history=None):
```

```
    if history == None:
        history = []
    history.append(value)
    return history
```

After (Fixing E711 Violation; is instead of ==)

```
~/ece444-pra7-starter main* ↑
.venv ❯ cat calculator.py
def add(a, b):
    return a + b


def record(value, history=None):
    if history is None:
        history = []
    history.append(value)
    return history

~/ece444-pra7-starter main* ↑
.venv ❯ git diff
diff --git a/calculator.py b/calculator.py
index 45f3f19..745022b 100644
--- a/calculator.py
+++ b/calculator.py
@@ -3,7 +3,7 @@ def add(a, b):


 def record(value, history=None):
-    if history == None:
+    if history is None:
         history = []
     history.append(value)
     return history

~/ece444-pra7-starter main* ↑
.venv ❯ python -m ruff check calculator.py

All checks passed!
```

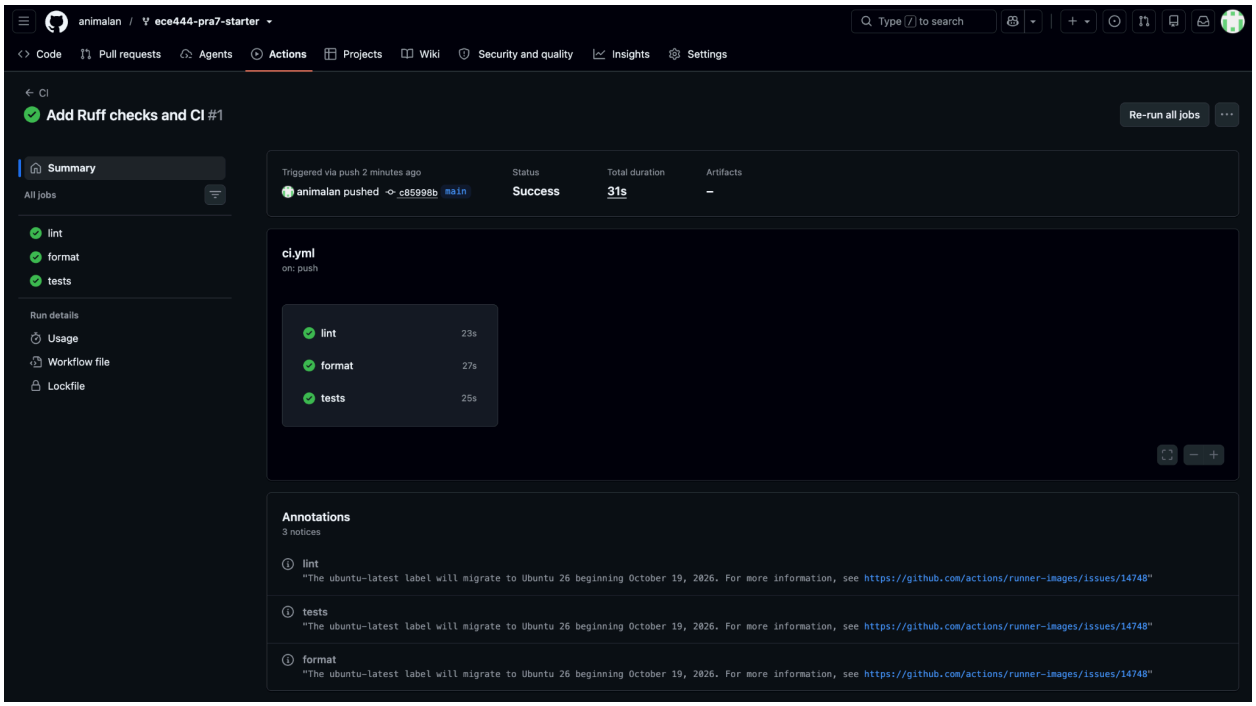
```
def add(a, b):
    return a + b



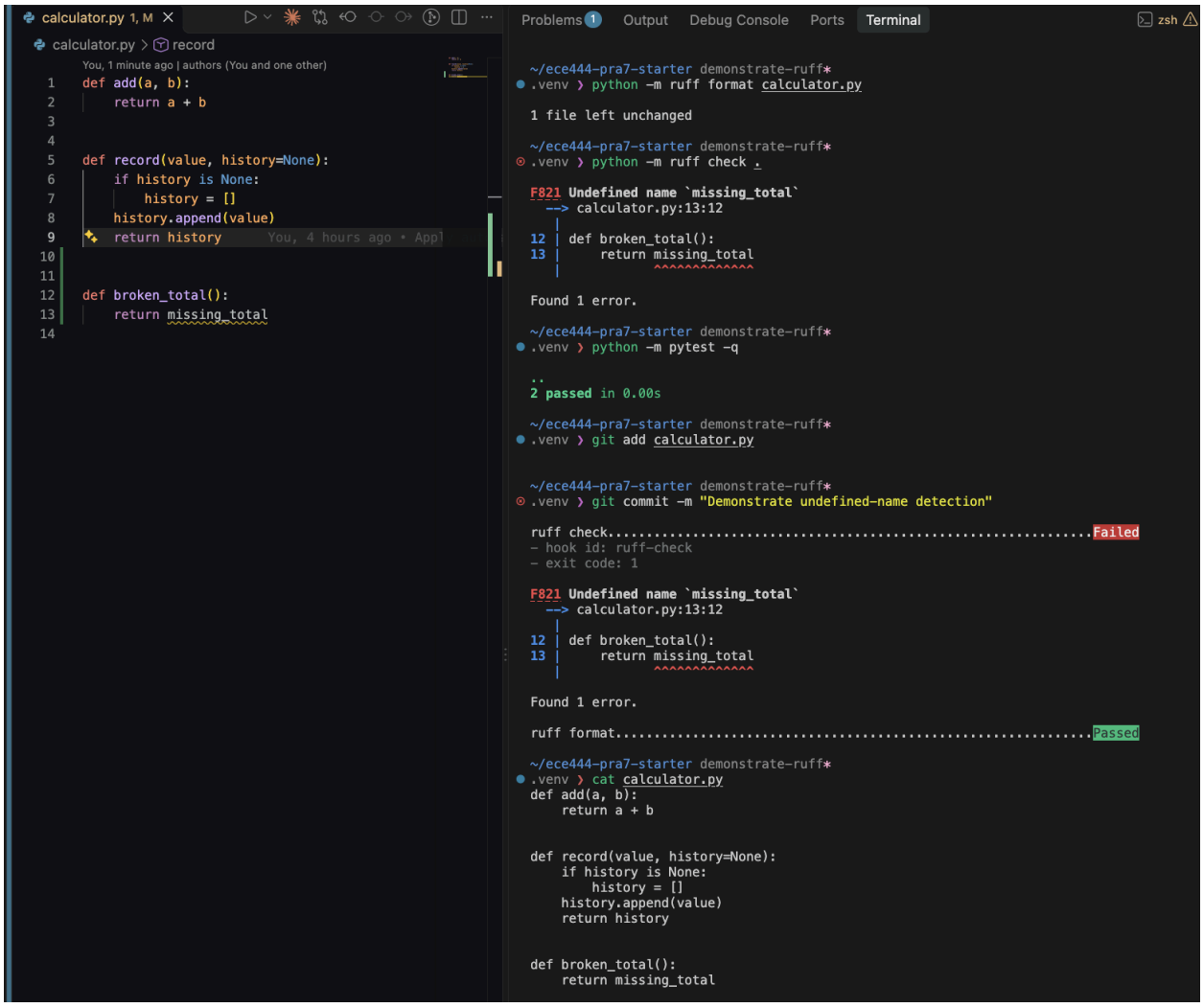
def record(value, history=None):
    if history is None:
        history = []
    history.append(value)
    return history
```

Item 6: Passing CI run on main

Run #1: https://github.com/animalan/ece444-pra7-starter/actions/runs/42735026774

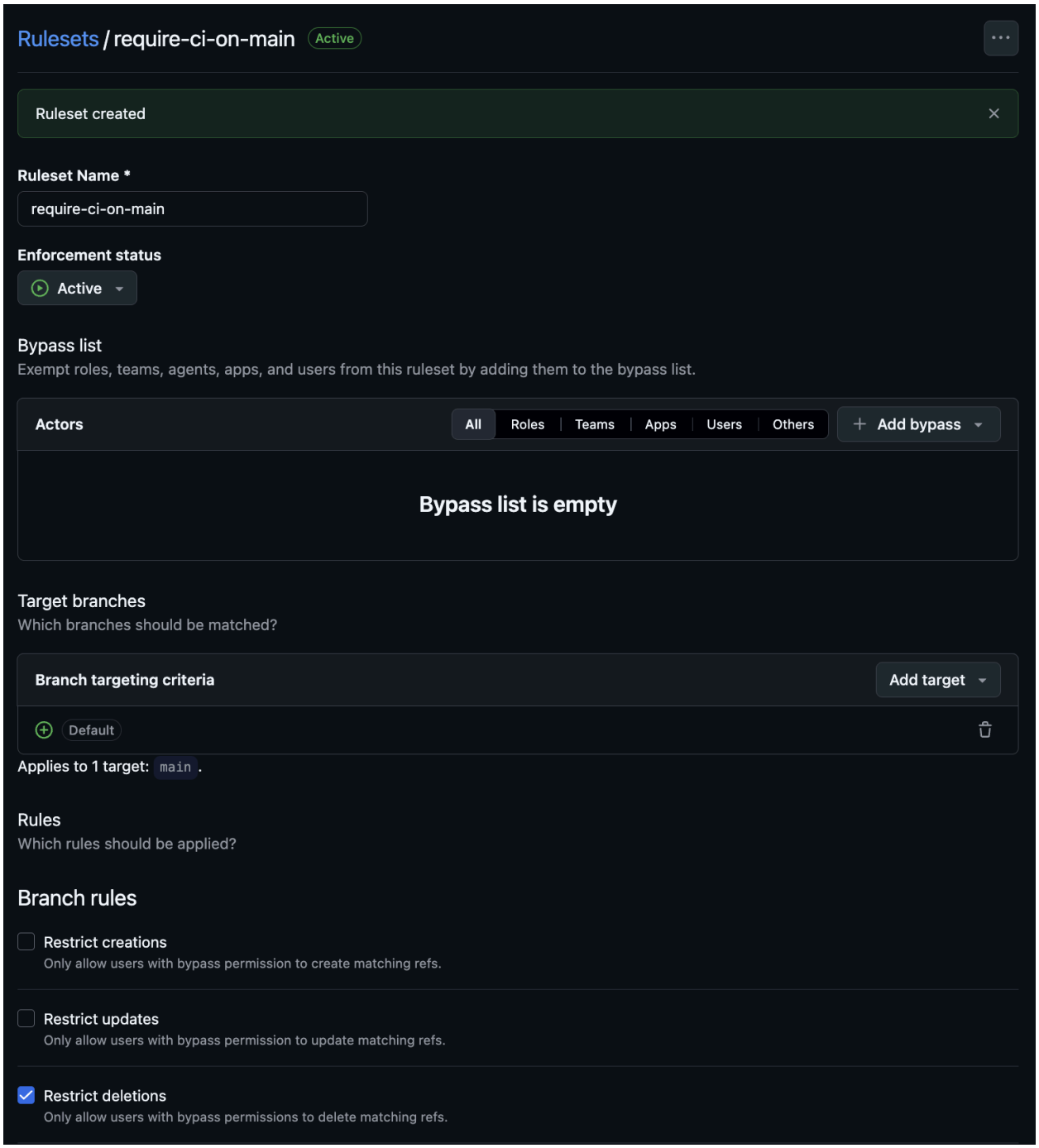


Item 7: Local failure of the defect



Item 8: Ruleset

Ruleset: https://github.com/animalan/ece444-pra7-starter/settings/rules/26908413



- [ ] **Require linear history**
  Prevent merge commits from being pushed to matching refs.

- [ ] **Require deployments to succeed**
  Choose which environments must be successfully deployed to before refs can be pushed into a ref that matches this rule.

- [ ] **Require signed commits**
  Commits pushed to matching refs must have verified signatures.

- [x] **Require a pull request before merging**
  Require all commits be made to a non-target branch and submitted via a pull request before they can be merged.

  Hide additional settings

  Required approvals

  0

  The number of approving reviews that are required before a pull request can be merged.

  - [ ] **Dismiss stale pull request approvals when new commits are pushed**
    New, reviewable commits pushed will dismiss previous pull request review approvals.

  - [ ] **Require review from specific teams**
    A collection of reviewers and associated file patterns. Each reviewer has a list of file patterns which determine the files that reviewer is required to review.

  - [ ] **Require review from Code Owners**
    Require an approving review in pull requests that modify files that have a designated code owner.

  - [ ] **Require approval of the most recent reviewable push**
    Whether the most recent reviewable push must be approved by someone other than the person who pushed it.

  - [ ] **Require conversation resolution before merging**
    All conversations on code must be resolved before a pull request can be merged.

  - [x] **Require an additional approval for unattributed Copilot pull requests**
    When Copilot opens a pull request without a human collaborator, require one more approving review if a non-zero approval count is required.

  Allowed merge methods

  Merge, Squash, Rebase

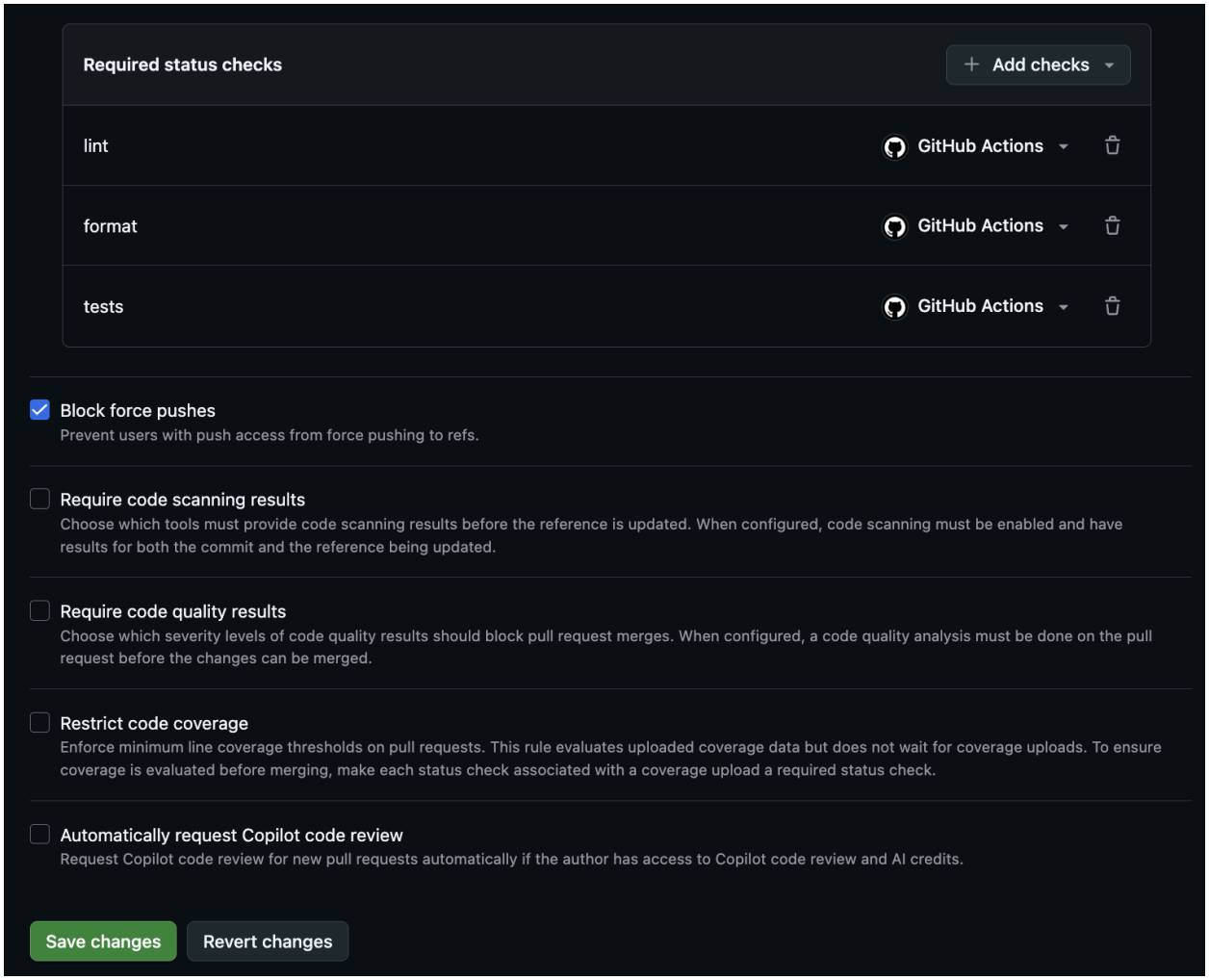
  When merging pull requests, you can allow any combination of merge commits, squashing, or rebasing. At least one option must be enabled.

- [x] **Require status checks to pass**
  Choose which status checks must pass before the ref is updated. When enabled, commits must first be pushed to another ref where the checks pass.

  Hide additional settings

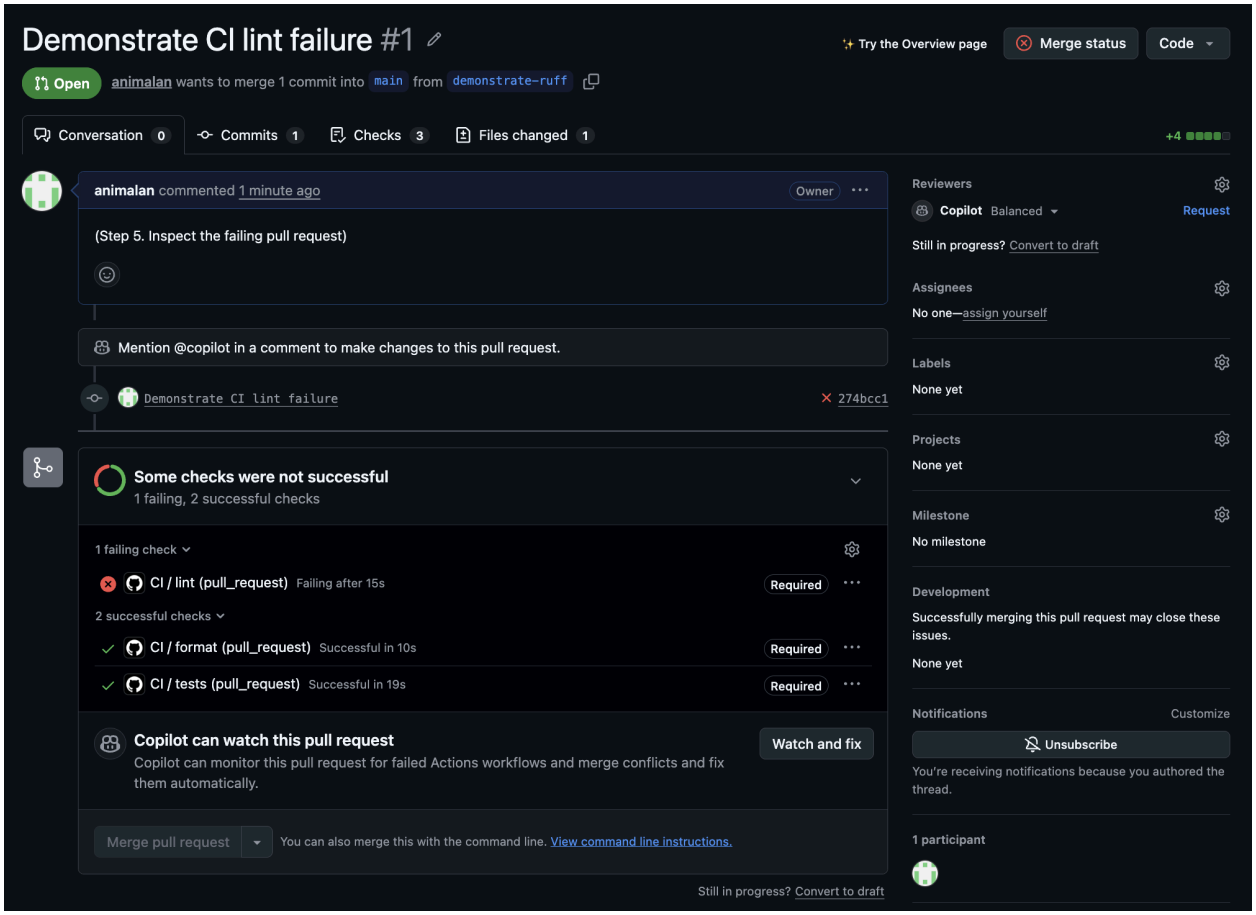
  - [ ] **Require branches to be up to date before merging**
    Whether pull requests targeting a matching branch must be tested with the latest code. This setting will not take effect unless at least one status check is enabled.

  - [ ] **Do not require status checks on creation**
    Allow repositories and branches to be created if a check would otherwise prohibit it.

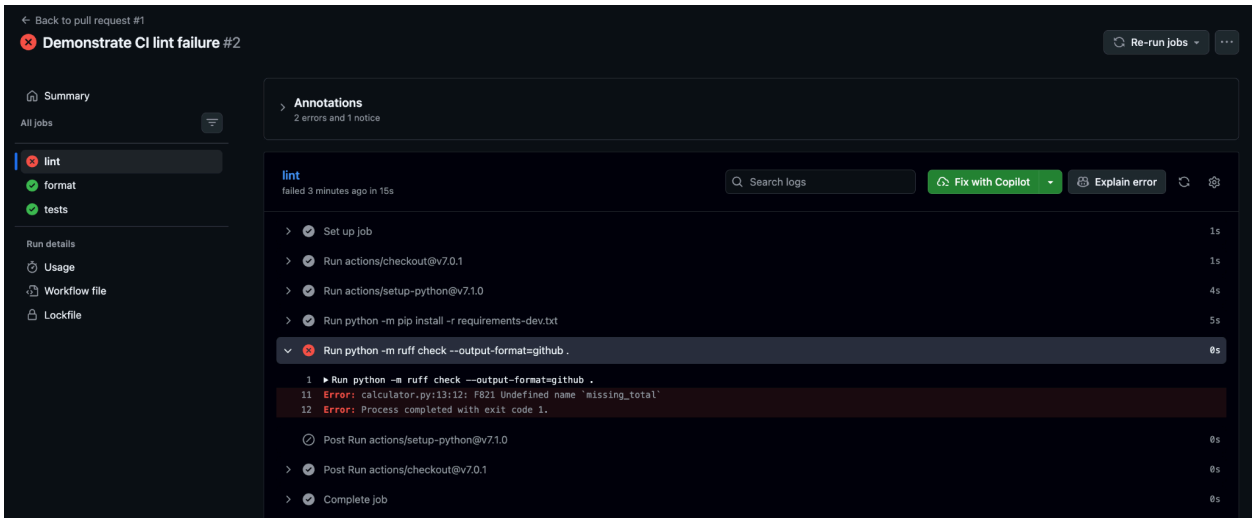

Required status checks
Add checks
lint
GitHub Actions
format
GitHub Actions
tests
GitHub Actions
Block force pushes
Prevent users with push access from force pushing to refs.
Require code scanning results
Choose which tools must provide code scanning results before the reference is updated. When configured, code scanning must be enabled and have results for both the commit and the reference being updated.
Require code quality results
Choose which severity levels of code quality results should block pull request merges. When configured, a code quality analysis must be done on the pull request before the changes can be merged.
Restrict code coverage
Enforce minimum line coverage thresholds on pull requests. This rule evaluates uploaded coverage data but does not wait for coverage uploads. To ensure coverage is evaluated before merging, make each status check associated with a coverage upload a required status check.
Automatically request Copilot code review
Request Copilot code review for new pull requests automatically if the author has access to Copilot code review and AI credits.
Save changes
Revert changes

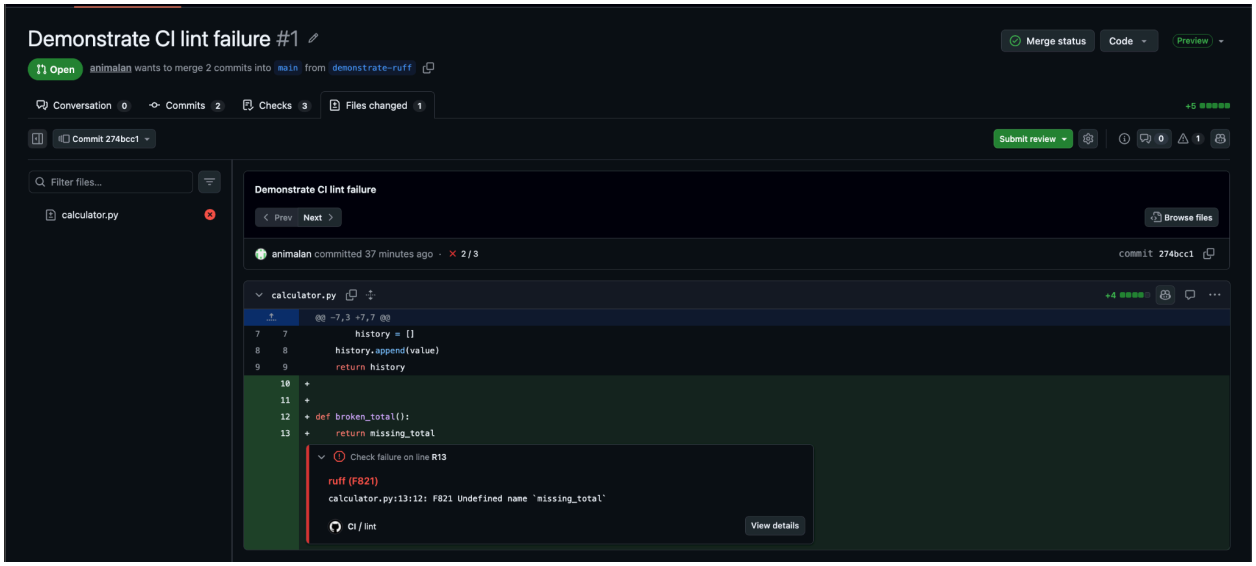
Item 9: Failing pull request

PR summary:



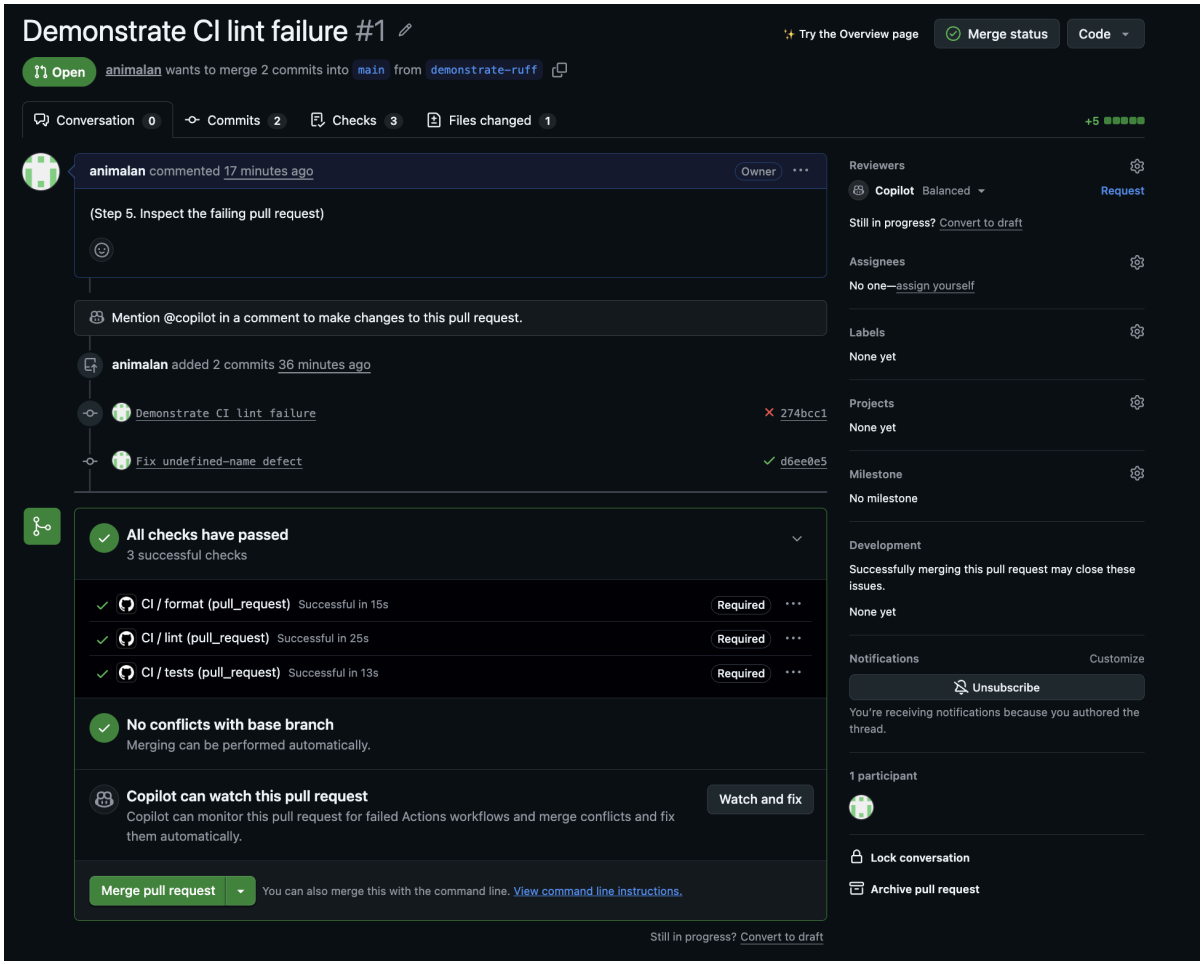
lint error:




Demonstrate CI lint failure #1
Merge status
Code
Preview
Open
animalan wants to merge 2 commits into main from demonstrate-ruff
Conversation 0
Commits 2
Checks 3
Files changed 1
+5
Commit 274bcc1
Submit review
Filter files...
calculator.py
Demonstrate CI lint failure
Prev Next
Browse files
animalan committed 37 minutes ago · 2 / 3
commit 274bcc1
calculator.py
+4
@@ -7,3 +7,7 @@
history = []
history.append(value)
return history
def broken_total():
return missing_total
Check failure on line R13
ruff (F821)
calculator.py:13:12: F821 Undefined name `missing_total`
CI / lint
View details

Item 10: Repaired pull request

PR: https://github.com/animalan/ece444-pra7-starter/pull/1




```python
def add(a, b):
    return a + b


def record(value, history=None):
    if history is None:
        history = []
    history.append(value)
    return history


def broken_total():
    missing_total = 0
    return missing_total
```